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The Microeconomics of the Space Market

Abstract

The economics of space exploration are rapidly evolving, currently driven by private companies like SpaceX and Blue Origin as opposed to the governments of the United States and Soviet Union in the past. The unregulated expansion of private ventures has spurred innovation but also raised concerns about orbital debris and unequal resource access. Regulatory capture poses a risk, necessitating robust, transparent frameworks to ensure fair competition and prevent monopolies. Corporate specialization is predicted to create an oligopolistic structure, akin to the airline industry, with a small number of firms focusing on markets like asteroid mining and space tourism. Lowering barriers to entry is crucial for fostering small businesses on other planets (which are crucial due to their prevention of a monopsonistic space labor market), but space debris remains as a significant obstacle that could lead to a “tragedy of the commons” scenario. Space exploration will become more economically attractive as Earth's resources dwindle thereby requiring replenishment from off-planet sources, which emphasizes the importance of early regulation and competition to support a sustainable multi-planetary economy that we can rely on. This research paper provides a comprehensive analysis of the microeconomic implications of space exploration, emphasizing the need for balanced regulation and the fostering of competition to ensure equitable development of a space market that balances regulation with innovation.

Introduction

The fascination with space exploration can be traced back to significant historical events such as the Apollo-Soyuz international space mission in 1975. The public sector (governmental) collaboration between the United States and the Soviet Union during this mission captivated the world, demonstrating the potential for international cooperation in space exploration. However, recent developments led by private companies like SpaceX have disrupted the previously government-dominated space sector by deploying satellites for both private and public entities. These advancements have generated considerable interest in the economic dynamics of the space market. The newly opened space market presents a unique opportunity to study the creation of a market and its functioning under new regulations (which are largely yet to be created). Understanding the economic principles underlying this emerging market is not only intellectually intriguing but also crucial for comprehending the future development of society itself.

Throughout history, societal advancements of similar magnitude have been accompanied by corresponding economic shifts. The flow of known and newly discovered natural resources from the Americas to Europe during colonization enriched certain individuals and businesses (along with entire countries, which is out of the scope of this paper). One might wonder if space colonization, involving the extraction of resources from asteroids, moons, and planets, will result in analogous economic outcomes. Additionally, the concept of harnessing solar energy from outer space and transmitting it to Earth could be compared to the transmission of newly discovered petroleum from the Middle East to Western economies.

Exploring these potential economic parallels will shed light on the transformative power of space activities.

Despite the growing interest in the economics of space exploration, the existing body of literature remains limited, primarily focusing on specific aspects such as asteroid mining. This research paper aims to provide a microeconomic analysis of the space market's expected evolution concerning the private sector as well as governmental regulation.

1. Unregulated Expansion

It is inherently challenging to predict the future, but it can be assumed with a certain degree of confidence that the private sector will continue to expand its presence in the space industry until it faces limitations imposed by government regulations. Companies like SpaceX have been at the forefront of driving innovation and accessibility in space exploration. For instance, SpaceX has recently introduced rideshare services, as indicated on their official website¹. This offering directly appeals to other companies by significantly reducing the barriers to entry for launching satellites into orbit. By providing cost-effective options for small satellite deployment, SpaceX has opened up new opportunities for various industries and organizations to access space.

Furthermore, SpaceX's endeavors extend beyond satellite deployment. The company has also made significant progress in its human spaceflight program. Through extensive testing and development, SpaceX aims to enable human travel beyond Earth's limits, with the ultimate goal of making life multiplanetary². Their human spaceflight program represents a clear

¹ <https://www.spacex.com/rideshare>

² <https://www.spacex.com/humanspaceflight/>

indication of the company's intention to extend the benefits of space exploration and colonization to private individuals and potentially even tourists.

While these developments are exciting and hold immense potential for scientific research, technological advancements, and commercial applications, they also raise concerns regarding unregulated expansion. Without proper oversight and regulations, the proliferation of private space exploration could lead to various challenges and risks. These may include issues related to orbital debris, interference with existing satellite networks, and unequal access to space resources.

As the private sector continues to push the boundaries of space exploration, it becomes increasingly important for governments and international bodies to collaborate in establishing clear guidelines and regulations. However, most of the standards introduced by SpaceX, the leading private space company, will likely remain in future markets as a side effect of first mover advantage. So, if the private sector remains a part of the pipeline towards satellite deployment, what happens next?

2. Regulatory Capture

As mentioned in Todd Sandler and William Schultze's paper *The Economics of Outer Space*, regulation in outer space is crucial. The United Nations passed the Outer Space Treaty³, preventing neo-colonization of and military activity in space. This attempts to prevent the exploitation of space due to its status as a common property good. It also emphasizes the principle of common heritage of mankind, stating that outer space and celestial bodies should

³ https://www.unoosa.org/pdf/gares/ARES_21_2222E.pdf

be explored and utilized for the benefit of all countries and humanity as a whole. However, as the space market evolves and private entities become increasingly involved, the risk of regulatory capture becomes more pronounced.

Private companies operating in space, such as SpaceX and Blue Origin, have significant resources and capabilities that can influence regulatory processes. The environment of space and the nature of technological advancements lends itself to the creation of natural monopolies, as Sandler and Schultz point out: "... in many cases the technology for outer space exploitation shows decreasing cost characteristics, consistent with natural monopoly resulting from scale economies. While exploitation to this point has been principally by government monopoly, this may not be the case in the future..." (Sandler and Schultz, 1981).

These powerful companies may then seek to shape regulations in their favor, potentially limiting competition or avoiding stringent oversight. Once regulatory capture occurs, the expansion of the space market will be similar to that of the colonization of the Americas, leading to an immediate exploitation of resources, a firm foothold in outer space, and an end result similar to the culmination of North America's colonization, thereby creating a splinter group which has control over Earth's only escape route. This can clearly lead to a problem if Earth's natural resources expire and humanity must turn to the stars, and the extraterrestrial resources that we once counted on are controlled by a monopoly.

To prevent regulatory capture in the space market, it is essential to establish robust and transparent regulatory frameworks. Governments and international bodies need to ensure that regulatory agencies have sufficient independence, resources, and expertise to effectively

oversee space activities. Conflict-of-interest rules and mechanisms to prevent undue influence should be in place to maintain the integrity of the regulatory process.

Additionally, fostering competition and diversity within the space industry can help mitigate the risks of regulatory capture. By encouraging multiple players and enabling smaller companies to enter the market, competition can provide a check on the influence of dominant players and promote innovation. Governments can support emerging companies through grants, subsidies, or preferential treatment in procurement processes, allowing smaller businesses to thrive and contribute to the overall development of the space market.

Furthermore, international cooperation and coordination are crucial in addressing regulatory capture in a global context. Given that space activities transcend national boundaries, collaboration between nations is necessary to establish harmonized regulations and standards. International bodies such as the United Nations Office for Outer Space Affairs (UNOOSA) and the International Telecommunication Union (ITU) can play a key role in facilitating dialogue and cooperation among countries to ensure that regulations align with the principles of equity, sustainability, and the peaceful use of outer space. Once these regulations are in place though, companies will still try to specialize in certain areas of the market to gain a competitive edge.

3. Corporate Specialization

The early space market is expected to be oligopolistic, with numerous major players in the private and public sectors and high barriers to entry. To prevent collaboration and the monopolization of outer space as a private commodity, regulations will be necessary. However,

these regulations will not immediately transform the market into a monopolistically competitive state. It will take several decades for the barriers to entry into outer space to significantly decrease. In a regulated oligopoly, each firm will have some product differentiation and pricing power. This competition will resemble the current airline market on Earth where there are high barriers to entry in the form of high startup costs and limited availability of takeoff and landing slots, along with unique opportunities in various market segments directed towards specific target demographics.

Using the analogy of the current Earth airline market, the competitors in the space market can be compared to Virgin Atlantic or Etihad Airways, known for their high prices and exclusive seats. To expand their individual demand, other space-faring firms may adopt a strategy similar to that of American Airlines, offering cheaper flights but with reduced onboard quality. On Earth, this strategy has proven effective, with mid-tier airlines like AA, United, and Delta holding a relative majority of around 16-17% each, according to the Bureau of Transportation Statistics⁴. It is surprising that Southwest, a value airline offering even lower prices but further reduced onboard quality, maintains a competitive market share of 17.2%.

However, it is unlikely that the space market will offer a similar economy option for space travel for many decades. Instead, companies are more likely to expand their range of space services rather than focus on significantly reducing costs to outcompete others, as shown by the Matthew Weinzierl, a Harvard professor who researches taxation, in his paper *Space, the Final Economic Frontier*, wherein he discusses the commercialization of space and nearly 40

⁴ <https://www.transtats.bts.gov/>

distinct firms and their engagement in the commercial industry (Weinzierl 6). Let us now analyze some of the potential specialized services that space firms could offer.

Asteroid Mining

Asteroid mining represents a significant area of corporate specialization within the space market. As the availability of resources on Earth becomes increasingly limited, the exploration and extraction of resources from asteroids hold great promise for meeting future demands. As Weinzierl points out, “A number of private companies are interested in mining asteroids for precious metals, in-space manufacturing inputs, habitat materials, and (perhaps most likely) water” (Weinzierl 16). Asteroids are rich in valuable minerals such as platinum, gold, and rare earth elements, which are essential for various industries, including electronics, aerospace, and renewable energy, and the potential for discovering an extraterrestrial source of water will of course be very sought after. Corporate entities specializing in asteroid mining have emerged, aiming to capitalize on these resources. These companies invest in advanced technologies and spacecraft capable of reaching and extracting resources from asteroids. Robotic mining missions are envisioned, where automated systems are deployed to identify, extract, and process valuable materials. This would involve utilizing technologies such as drilling, blasting, and material separation techniques to extract and refine resources from asteroids.

Space Tourism

Corporate specialization in space tourism has emerged as a promising avenue for companies seeking to tap into the growing demand for extraterrestrial experiences, similar to the current airline market and its popular use for vacation trips. With the advancements in

space technology, several companies have begun focusing on providing unique and tailored space tourism services, such as Virgin Atlantic's extraterrestrial counterpart, Virgin Galactic (Weinzierl 6). These specialized firms aim to offer unforgettable space travel experiences⁵, likely catering to higher-end customer's with similarly high budgets.

The development of space tourism will create a diverse range of jobs. This includes aerospace engineers, spacecraft designers, pilots, astronauts, hospitality staff for spaceports and hotels, tour guides, and support personnel. The pursuit of space tourism will further drive innovation in various technological areas. Companies will invest in spacecraft design and manufacturing, propulsion systems, life support systems, and safety protocols. These advancements will have ripple effects, leading to improvements in satellite technology, materials science, renewable energy, and telecommunications, in addition to the robust supply chain needed to support space travel that will have to be created and maintained. It is therefore likely that most of the advancement in space travel will stem from competition in the tourism industry and an incentive to optimize more than the competition, leading to the most effective and useful form of corporate specialization.

Space Colonization

Space colonization represents a long-term vision for the expansion of human civilization beyond Earth. Corporate specialization in space colonization involves the establishment of sustainable habitats and infrastructure on other celestial bodies, such as the Moon, Mars, or even large asteroids. Such habitats would support human life, enabling individuals to reside, work, and conduct research in extraterrestrial environments. Corporate entities specializing in

⁵ <https://brochure.virgingalactic.com/>

colonization would engage in the development of technologies for constructing and maintaining habitats, resource utilization, transportation systems, and life support systems. They would be responsible for the design and construction of modules, habitats, and infrastructure necessary for sustainable living and scientific exploration. These companies would also address challenges related to food production, waste management, energy generation, and healthcare in extraterrestrial environments.

The economic implications of colonization are significant, as it would require the establishment of land-based markets, trade networks, and governance structures to support the growing population and ensure the sustainability of the colonies. Corporate entities specializing in space colonization would play a vital role in shaping these economic systems and facilitating the growth and development of extraterrestrial communities.

4. Barrier Lowering and Small Businesses in Space

Returning to Matthew Weinzierl, the commercialization of space has been accepted by the government: “As the Shuttle program wound down, the primary channel by which NASA and the rising New Space sector came together to solve the space access problem—and thereby provide an example of how decentralization can work—was a set of public–private partnerships called Commercial Orbital Transportation Services (COTS)” (Weinzierl 8). The COTS project, as Weinzierl points out, was the first to break the private sector barrier to entry of commercial space when NASA risked \$500 million dollars in partnership with Orbital Sciences and SpaceX to create the Commercial Crew & Cargo Program Office, charged with the task of “stimulating commercial enterprise in space by asking American entrepreneurs to provide

innovative, cost-effective commercial cargo and crew transportation services to the [international] space station,” according to NASA’s COTS documentation⁶. It is clear, then, that the barrier of commercial space has been broken, albeit only slightly and by a collaboration of the biggest players in the market.

Returning to the analogy discussed in Section 3, the early phase of the airline industry was not unlike the current space travel dynamic. Most rides were quite harsh, and the wealthy customers did not tend to travel by plane at all. In fact, airlines mainly delivered mail for the government in order to make enough money to support their expansion.⁷ The space equivalent of this mail could be anything from space exploration tools, such as biosphere materials for colonization or mining tools for asteroid collection, to general satellites for government or even collegiate institutions, such as with NASA’s CubeSat initiative⁸.

Barrier lowering for the airline industry is a process that took nearly an entire century to reach the stable oligopoly state seen in the current market. It is, however, almost guaranteed to take more than a century for an equivalent steady state to be reached in the space market, simply because of the risks associated with space travel, such as colliding with the remnants of space travel attempts. Space debris has been tracked vigorously by NASA’s Orbital Debris Program Office and modelling tools show⁹ how the Earth’s atmosphere has become congested with space debris – no matter the size of the debris, they can still be greatly damaging at the speed of rockets.

⁶ <https://www.nasa.gov/wp-content/uploads/2016/08/sp-2014-617.pdf>

⁷ <https://airandspace.si.edu/explore/stories/evolution-commercial-flying-experience>

⁸ <https://www.nasa.gov/cubesat-launch-initiative-introduction/>

⁹ <https://orbitaldebris.jsc.nasa.gov/modeling/>

Space debris is a notably unique problem that does not have an equivalent in other Earth markets. The failed attempts of airplane travel lie far from the path that future planes will take... although even successful space launches can release debris¹⁰ from their stage separations or the thermal stress from passing through the atmosphere. This leads to an interesting paradigm for the amount of traffic that should be allowed into space at one time, and whether or not corporations will specialize into space-debris cleaning ships. The “tragedy of the space commons” is a term to describe the cycle from this negative externality (Chaddha 2010). The more companies start sending rockets to space without regulation on debris production and cleanup, the greater the likelihood of collisions, halting rocket travel. It seems that the barriers to entry are regulatory, financial, and physical in nature. This compound effect will dramatically increase the time it takes for commercial space expansion to occur, along with the relative lack of incentive.

As of now, the main incentive for space exploration is just that – exploring the opportunities that lie in outer space. The incentive, especially for space habitation and colonization, will not be present at least until Earth’s resources expire and life in a bubble on an arid planet is better than life on Earth. At this point, the entirety of Earth’s economy will have to switch to a planetary or multi-planetary system, and we then encounter the problem of small business in space.

Job opportunities will be far from limited once we are a multi-terrestrial species. To simply keep one inhabitable biosphere up to the new standard of living will require many workers. If we plan on living in space stations, maintaining artificial gravity and appropriate

¹⁰ <https://orbitaldebris.jsc.nasa.gov/faq/#>

living conditions will still be a challenge. The problem with this lifestyle, however, is that we would effectively have to restart this entire process of developing a new economy in order to have something comparable to small businesses in space – the breaking of barriers-to-entry by a small group of risk-takers, the creation of incentives to use this new small business, the lowering of barriers, and finally, the creation of other small businesses. This long process will become even harder when our center of habitation is – literally – up in the air. Unfortunately, this process is necessary in order to prevent the same monopolistic/regulatory capture from development of the space market from happening in the space labor market. In short, it is very likely that, if we inhabit space out of desperation, a monopsonistic labor market will develop in space and keep a tight rein on the possibility of a free market developing.

The only way that this small business expansion could happen, then, is by getting a head start on space travel and habitation. The sooner we arrive on the stars, the less we will have to rely on the stars to survive. Once there are options on different planets, a free market economy can truly thrive, wherein the labor demand is free to roam wherever the conditions are best.

5. Conclusion

As we stand on the brink of a new era in space exploration, this research underscores the profound economic implications and potential transformations within the space market. The burgeoning private sector's role in space exploration, exemplified by SpaceX's pioneering efforts, indicates a significant shift towards increased accessibility and innovation. However, the potential for unregulated expansion raises critical concerns about orbital debris, equitable access to resources, and the risk of monopolistic control. As private entities continue to push

boundaries, the establishment of robust regulatory frameworks by governments and international bodies becomes imperative. Corporate specialization in areas such as asteroid mining, space tourism, and space colonization illustrates the diverse opportunities within the space market. These specializations not only promise economic benefits but also drive technological advancements and job creation. However, the unique challenges of space, such as space debris, require innovative solutions and collaborative efforts to ensure the long-term viability of space activities. Finally, the lowering of barriers to entry and the fostering of small businesses in space will be essential for a thriving, competitive market. The proactive pursuit of space exploration and colonization, driven by curiosity rather than necessity, will be key to avoiding monopolistic tendencies and ensuring a diverse and vibrant economic landscape beyond Earth.

The future of the space market can be envisioned through several distinct frameworks of regulation and control. These frameworks can be broadly categorized into three possible ultimate states. The first is an authoritarian environment with strict regulations leading to effective control and sustainability with limited private sector influence, but a more centralized market structure and limited individual innovation and incentives. The second possible state is one of private sector domination, where private companies have significant control, driven by free market forces and providing innovation at the expense of a risk of monopolization and a detrimental neglect of environmental and sustainability interests. The final possible state is libertarian, with a chaotic lack of regulation. There would be high levels of free competition and innovation, with an extreme risk of unsustainable expansion and resource conflicts.

Each of these ultimate states presents a different pathway for the future of the space market. The future of space exploration hinges on our ability to balance innovation with regulation, competition with collaboration, and ambition with sustainability. In space, we must distribute communal, public, and private rights in order to unlock the potential of the space market and pave the way for a prosperous space economy that benefits all of humanity.

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